Medical Image Classification with ResNet-50

Lab Note

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2026-May

Prepared for Year 1-2 undergraduate students new to practical deep learning
With special thanks to Prof. Philip Yu for advice on pedagogical design.

What you will build
A complete transfer-learning pipeline that loads cervical spine X-ray images, trains a ResNet-50 classifier, evaluates class-level performance, and uses Grad-CAM to inspect where the model focuses.

Learning Objectives

1. Operate a GPU-powered notebook on the cloud.
2. Construct a PyTorch data pipeline with stratified splitting and medically-appropriate augmentation.
3. Apply transfer learning with a pretrained ResNet-50 and implement best-practice training (early stopping, LR scheduling).
4. Evaluate the model using accuracy, per-class precision/recall/F1, and a confusion matrix rather than relying on one headline metric.
5. Use Grad-CAM to check whether the model focuses on the cervical vertebrae rather than irrelevant image borders or artifacts.
6. Explain the practical limitations of a small medical dataset and propose one evidence-based improvement after the lab.

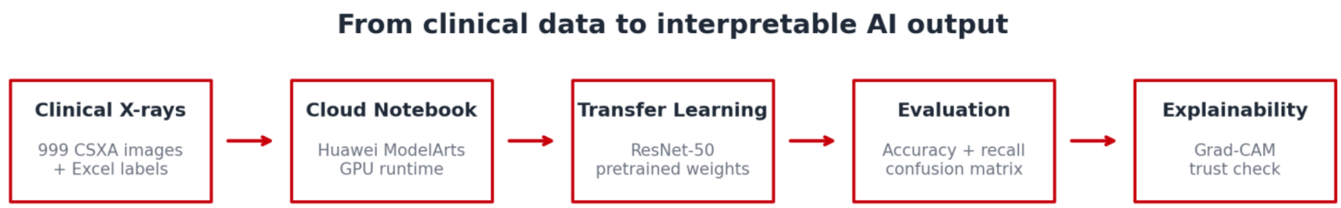
Safety and ethics note
This lab is for education and research training only. The model is not a diagnostic system, has not been clinically validated, and must not be used for patient care.

1. Why This Lab Matters

Many students and IT professionals spend long hours in front of digital screens, and neck discomfort is a common clinical complaint in this population. When cervical spine alignment needs to be assessed, X-ray imaging is one widely used clinical tool. In this lab, we use cervical spine X-rays to study how deep learning can support curvature classification.

We utilize the **Cervical Spine X-ray Atlas (CSXA)** dataset — 999 real, expert-annotated clinical images. The task is framed as a 4-class problem: **Lordotic (normal), Straight, Sigmoid, and Kyphotic**.

> **Key Concept: Authentic Learning vs. Toy Datasets**
> Most academic tutorials rely on sanitized datasets (MNIST). These fail to capture real-world clinical challenges: severe class imbalance, subtle inter-class visual boundaries, and label ambiguity between Straight and Sigmoid curvatures. The CSXA dataset exposes students to these authentic difficulties from day one.



Modern machine learning is rarely performed on local laptops. This lab standardizes the environment using Huawei Cloud, providing every student with an identical NVIDIA GPU. This mirrors real industry workflows and eliminates "works on my machine" friction.

2. Preparation: Notebook Setup

Huawei Cloud ModelArts is used because it gives the class a standardized GPU notebook environment. This removes a common teaching problem: different laptops, different drivers, different Python versions, and different results.

1. Open the notebook instance.
2. Select a PyTorch environment with GPU support.
3. Open JupyterLab and locate the provided Kaggle/ModelArts notebook.
4. Run the environment-check cell before loading data.

```
import torch
print('torch cuda available:', torch.cuda.is_available())
print('torch version:', torch.__version__)
!nvidia-smi
```

3. Dataset: Loading Images and Matching Labels

The notebook downloads the image folder and the Excel label file. Each image filename begins with a four-digit identifier. The Excel sheet contains the corresponding image number and curvature label. The first practical task is therefore not modeling - it is reliable data joining.

```
IMAGE_DIR = kagglehub.dataset_download('ddatad/cervical-x-ray')
LABEL_ROOT = kagglehub.dataset_download('ddatad/dataset')
LABEL_FILE = os.path.join(LABEL_ROOT, 'dataset.xlsx')

df = pd.read_excel(LABEL_FILE, engine='openpyxl')
df['Curvature'] = df['Curvature'].replace({4: 3})  # merge Sigmoid1 and Sigmoid2
```

The lab merges Sigmoid1 and Sigmoid2 into one Sigmoid class. This is a teaching design choice: the two subtypes are visually subtle.

We perform a stratified 80/20 train-validation split to preserve class proportion.

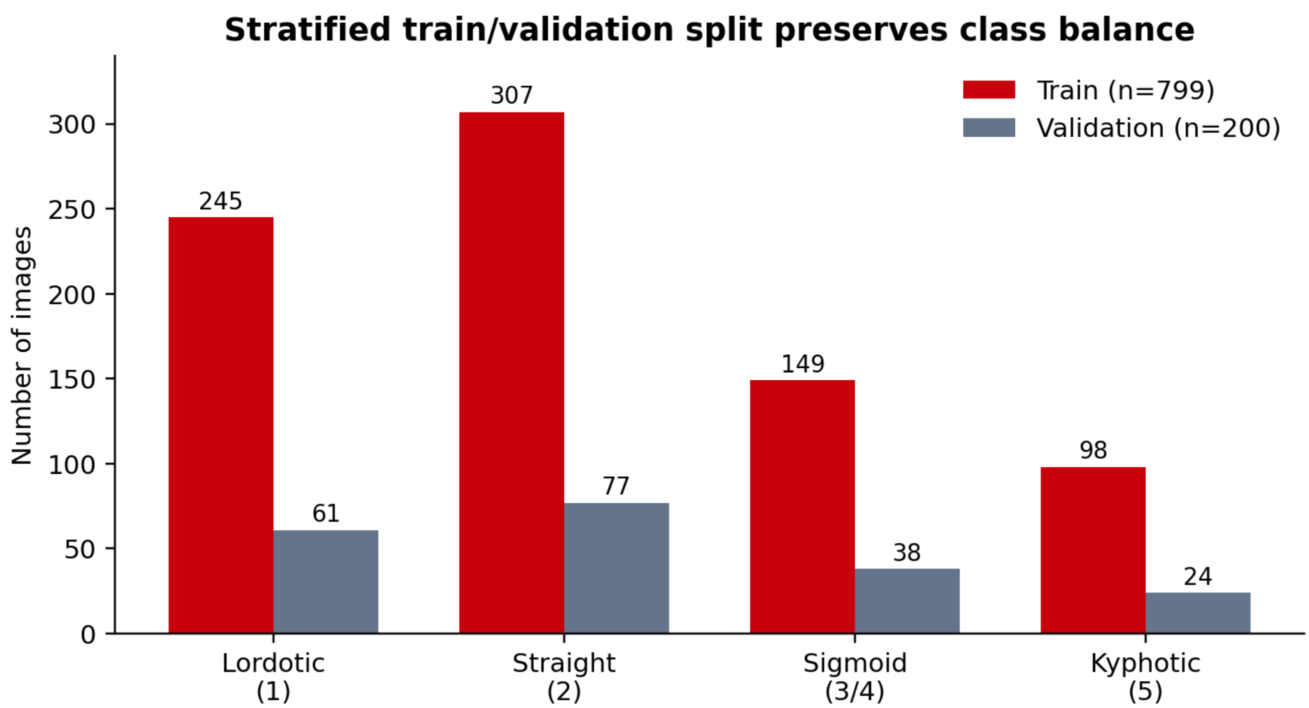


Figure 1: Stratified Train/Validation Split (Total 999 Images)

Straight is the majority class; Kyphotic is the minority. Stratification ensures both splits reflect this distribution.

4. Image Transforms: Gentle Augmentation, Not Random Distortion

Augmentation is used to reduce the model's sensitivity to small changes in patient positioning. However, medical augmentation must be conservative. We want to simulate plausible acquisition differences, not create anatomically unrealistic images.

Transform	Why do we use it	Practical caution
Resize + crop to 224x224	ResNet-50 expects a standard input size.	Avoid cropping away the cervical spine.
Random rotation ±10°	Simulates minor positioning differences.	Large rotations can create unrealistic anatomy.
Horizontal flip	Adds diversity in this teaching task.	In some medical tasks, left/right orientation matters; check with clinicians.
ImageNet normalization	Matches the scale expected by pretrained ResNet-50.	Do not remove it unless you retrain/retune carefully.

```
train_tfms = T.Compose([
    T.Resize((256, 256)),
    T.RandomResizedCrop(224, scale=(0.85, 1.0)),
    T.RandomHorizontalFlip(),
    T.RandomRotation(10),
    T.ToTensor(),
    T.Normalize([0.485,0.456,0.406], [0.229,0.224,0.225])
])
```

5. Model: Fine-Tuning ResNet-50

Training a deep convolutional neural network from scratch on 999 images would typically lead to overfitting. Transfer learning is the practical alternative: start with a model that has already learned general visual features from a large image dataset, then adapt it to the X-ray task.

Why ResNet Is Easier to Train： The Power of Skip Connections

Deep networks often suffer from the "vanishing gradient problem," in which learning signals diminish as they propagate backward, leading to early layers ceasing to learn.

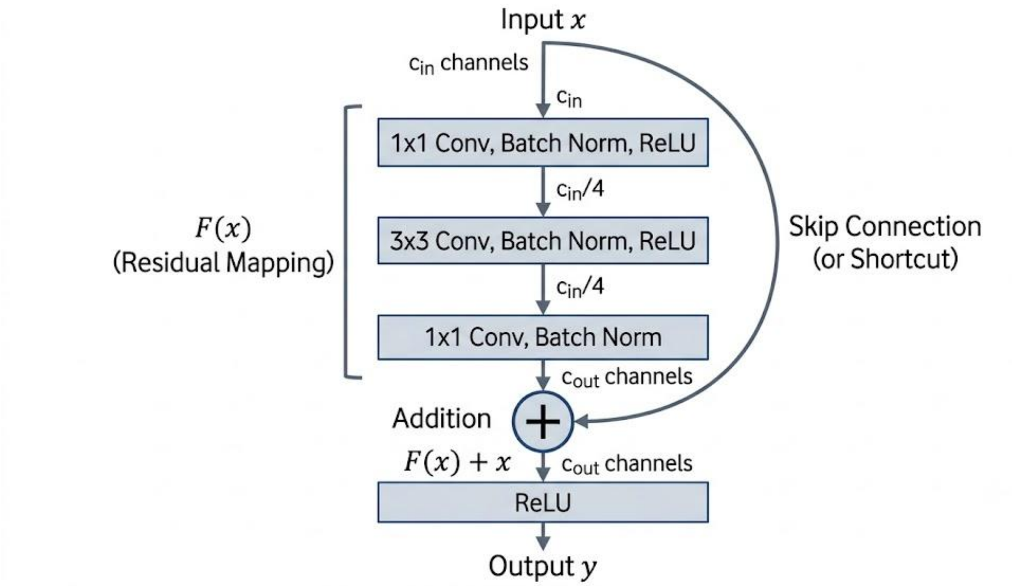


Figure 2: In ResNet-50, skip connections help information flow through deep networks

ResNet solves this mathematically with skip connections (See Figure 2). By routing the input x directly around a layer block and adding it to the output, the network learns a residual mapping: y = F(x) + x. During backpropagation, the gradient of this output with respect to the input is:

$$\frac{\partial_y}{\partial_x} = \frac{\partial F(x)}{\partial_x} + 1$$

Even if the layers are so deep that their internal gradient ($\frac{\partial F(x)}{\partial_x}$) approaches zero, the constant + 1 ensures the total gradient never vanishes. This creates an uninterrupted "gradient highway," enabling deep models to optimize safely without training collapse.

The pretrained model already knows useful low-level visual features such as edges, contrast changes, and texture patterns. Fine-tuning adapts those features to the medical image task with fewer examples.

```
model = models.resnet50(weights=models.ResNet50_Weights.IMAGENET1K_V1)
model.fc = nn.Linear(model.fc.in_features, num_classes)
model = model.to(device)

criterion = nn.CrossEntropyLoss()
optimizer = torch.optim.Adam(model.parameters(), lr=1e-4, weight_decay=1e-4)
scheduler = torch.optim.lr_scheduler.ReduceLROnPlateau(
    optimizer, mode='min', factor=0.5, patience=2)
```

We use a small learning rate (1e-4) and weight decay (1e-4). This protects the valuable low-level features (edges, bone textures) already learned by the ImageNet-pretrained backbone while allowing gentle adaptation to X-ray data.

6. Training: What to Watch During the Run

During training, students should not only wait for the final score. They should monitor whether training loss and validation loss move together. When training improves, but validation gets worse, the model is starting to memorize the training set.

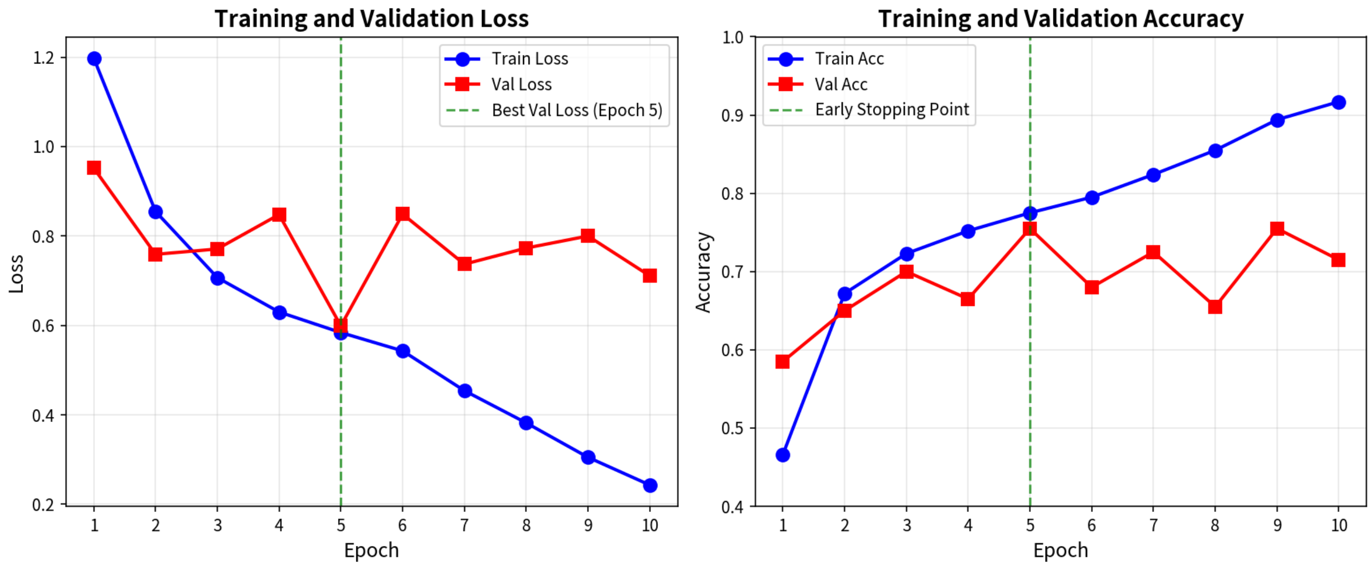


Figure 3: Typical training run: validation loss reaches its best point before later overfitting

Validation loss reaches its minimum at Epoch 5. Continued training causes overfitting — early stopping automatically restores the best checkpoint.

Signal	Healthy pattern	Warning pattern
Training loss	Generally decreases over epochs.	Does not decrease: learning rate too low, data issue, or model not training.
Validation loss	Decreases, then stabilizes.	Increases while training loss decreases: overfitting.
Validation accuracy	Improves and then plateaus.	Large jumps: the validation set may be small, or class performance may be unstable.

7. Holistic Evaluation: Global Accuracy Is Not Enough

In clinical AI, global accuracy is a dangerously incomplete metric. We must examine performance at the per-class level.

> **Key Concept: The Sigmoid Failure Case**
>
> While Straight achieves 88% recall and Kyphotic 83%, the Sigmoid class collapses to only 29% recall. Two factors cause this: (1) genuine visual ambiguity — the physiological boundary between Straight and Sigmoid is subtle even to expert radiologists, and (2) data imbalance — the model is biased toward majority classes. Global accuracy masks this critical weakness.

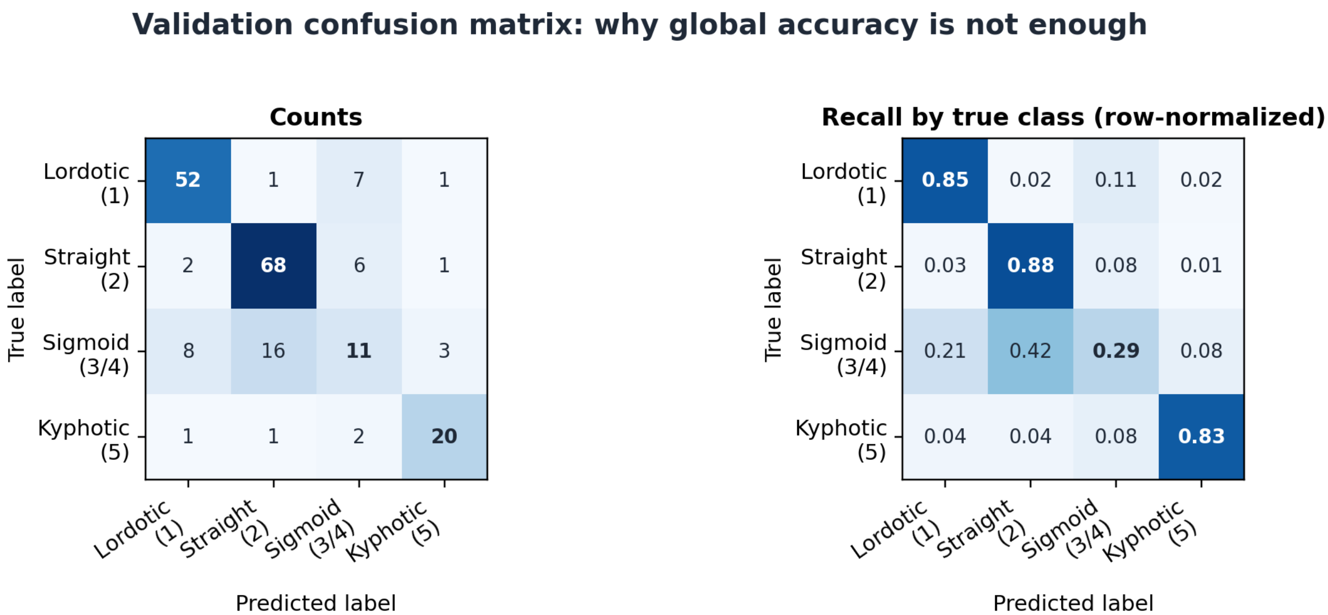


Figure 4: Validation confusion matrix for the baseline ResNet-50 model. Left: raw counts. Right: row-normalized values, which can be read as recall for each true class.

Row-normalized values = recall per class. Notice the severe under-performance on Sigmoid (only 29% of true Sigmoid cases correctly identified).

The model achieves 75.5% validation accuracy, but the row-normalized matrix shows the real issue: Lordotic, Straight, and Kyphotic have high recall, while Sigmoid is much harder.

8. Grad-CAM: Inspecting Where the Model Looks

Deploying black-box models in healthcare carries ethical and safety risks. We must verify that the network attends to the same anatomical structures a radiologist would examine. We therefore apply **Gradient-weighted Class Activation Mapping (Grad-CAM)** to the final convolutional block.

```
heatmap = grad_cam(
    model,
    img.unsqueeze(0).to(device),
    class_idx=pred,
    target_layer='layer4')
```

Grad-CAM computes the gradient of the target class score with respect to the feature maps, then produces a spatial heatmap overlaid on the original X-ray. A trustworthy model exhibits strong activation (red/yellow zones) localized precisely over the cervical vertebrae, ignoring irrelevant background or artifacts.

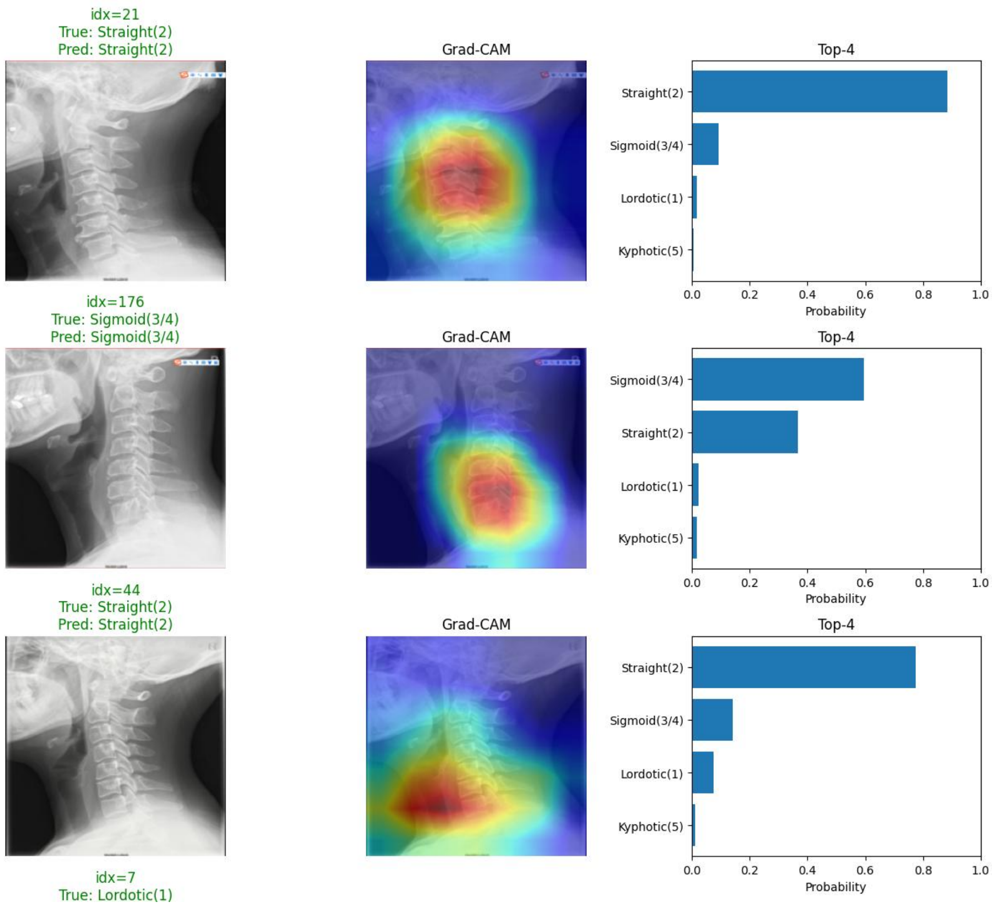


Figure 5: Grad-CAM is used as a sanity check: the activation should fall over the cervical vertebrae, not over image borders, labels, or background artifacts.

Grad-CAM observation	Interpretation
Activation concentrated on the cervical vertebrae.	Good sign: the model is looking at anatomically relevant regions.
Activation on borders, labels, background, or artifacts	Warning sign: the model may be using shortcuts unrelated to curvature.
Correct location but wrong prediction	The model may attend to anatomy but still extract insufficient or misleading features.

⚠️ The Grad-Cam Heatmap is a sanity check, not a proof of clinical reasoning. It shows where the model is looking, but it cannot tell you what the model is concluding.

9. In-Class Deliverables

Submit the following before leaving the lab

Done	Task
[ ]	Screenshot of train and validation class distributions.
[ ]	Screenshot learning curve figure or printed epoch logs.
[ ]	Screenshot the Confusion matrix and classification report.
[ ]	Pick one Grad-CAM example with your interpretation.
[ ]	Identify the weakest class and explain one possible reason.

10. After-Class Inquiry Task

The baseline model struggles with the Sigmoid class. Your after-class task is to act as the lead engineer and propose one evidence-based improvement. You may consult a Generative AI assistant such as ChatGPT or DeepSeek for possible strategies, such as class-weighted loss, oversampling, stronger augmentation, or threshold calibration. However, you must critique the suggestion before using it. Select one strategy, implement it, retrain the model, and compare the result with the baseline.

GenAI use rule

You may consult ChatGPT, DeepSeek, or another assistant for ideas. However, you must critique the suggestion before using it. Do not submit unexamined copy-pasted code.

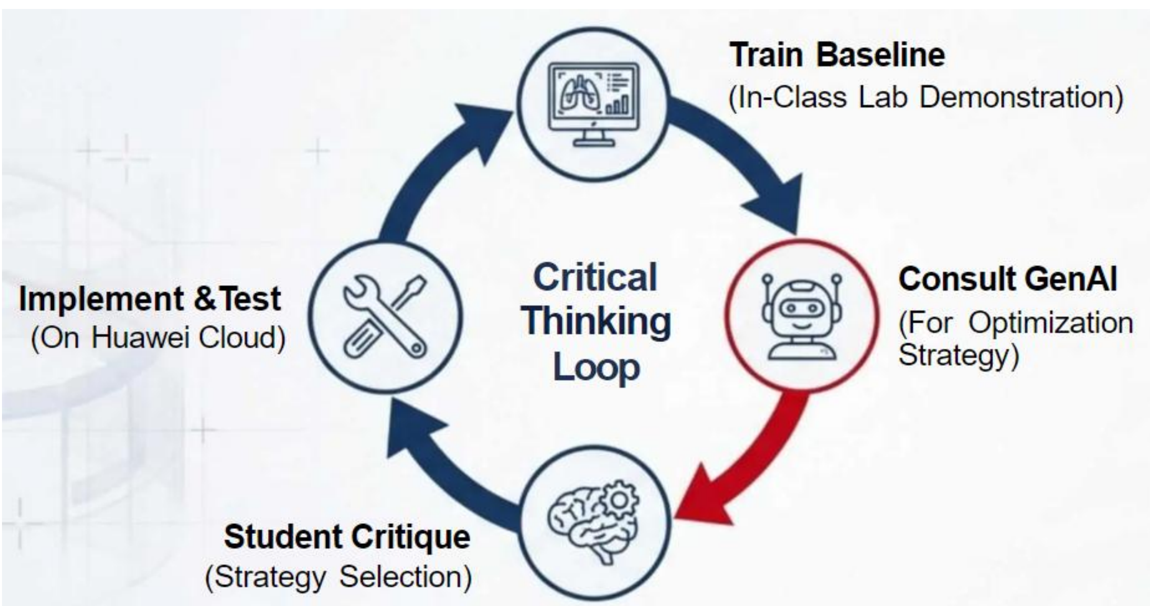


Figure 6. The Critical Thinking Loop (Human + GenAI as Reflective Partner)

Further Reading

Papers for ResNet & Grad-CAM

1. *He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. CVPR. https://arxiv.org/pdf/1512.03385.pdf*
2. *Selvaraju, R. R., et al. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. ICCV. https://arxiv.org/pdf/1610.02391.pdf*

Online Resources for Transfer Learning & ResNet

1. *PyTorch Transfer Learning Tutorial https://pytorch.org/tutorials/beginner/transfer_learning_tutorial.html*
2. *Dive into Deep Learning — ResNet Chapter (Free textbook) https://d2l.ai/chapter_convolutional-modern/resnet.html*

CS231n: Convolutional Neural Networks for Visual Recognition (Stanford) https://cs231n.stanford.edu/

Lab Code & Data

1. *Full lab code repository https://github.com/098765d/Huawei-ICT-Teaching-Competition-2025-26*
2. *CSXA Dataset on Kaggle https://www.kaggle.com/ddatad/cervical-x-ray*